

A. Effective opacity

con_o

$$\text{con_o} = (\text{conic}.x, \text{conic}.y, \text{conic}.z, o_i \cdot \text{coef}_i)$$

coef_i: low-pass compensation from the +0.1 dilation



B. Tile filtering

duplicateToTilesTouched

Ellipse where $\alpha \geq 1/255$; threshold $t_{thresh} = \text{mult} \cdot 2 \ln(255 o_i \text{coef}_i)$

Tiles touched = 0 $\Rightarrow$ Gaussian removed from the pipeline



C. Normal from quaternion

$q = (r, x, y, z)$

$$\mathbf{n}_i = (2(xz + ry), 2(yz - rx), 1 - 2(x^2 + y^2))$$

Third column of $R(q)$: local z -axis after rotation (approximate normal of a flat Gaussian)